



FATIMA JINNAH WOMEN UNIVERSITY

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Section:

5A

CLOUD COMPUTING

Lab No.14

Initial Setup

The image shows two screenshots related to the initial setup of a Terraform project.

Top Screenshot: GitHub Repository View

The repository is named **terraform_machine** and is forked from [WaqarSaleem97/terraform_machine](#). It is a public repository with 1 branch (main) and 0 tags. The repository was created 3 days ago by **WaqarSaleem97** and has 5 commits. The files listed are:

File	Added	Time
modules	Add files via upload	3 days ago
.gitignore	Add files via upload	3 days ago
README.md	Add files via upload	3 days ago
locals.tf	Add files via upload	3 days ago
main.tf	Add files via upload	3 days ago
outputs.tf	Add files via upload	3 days ago

The right sidebar shows the repository's metadata, including no description, website, or topics provided, and no releases or packages published.

Bottom Screenshot: VS Code Codespace Environment

The VS Code interface shows the **EXPLORER** view with the following files:

- TERRAFORM_MACHINE [CODESPACES]
- modules
- .gitignore
- locals.tf
- main.tf
- outputs.tf
- README.md
- variables.tf

The **TERMINAL** view shows the following output:

```
- Want to use a custom image instead? Learn more here: https://aka.ms/configure-codespace
To explore VS Code to its fullest, search using the Command Palette (Cmd/Ctrl + Shift + P or F1).
@23-22411-027-max → /workspaces/terraform_machine (main) $
```

The bottom right corner of the terminal shows a prompt to open chat: `Open chat [Ctrl]+[I]. Start typing to dismiss.`

🔍 To explore VS Code to its fullest, search using the Command Palette (Cmd/Ctrl + Shift + P or F1).

- @23-22411-027-max → /workspaces/terraform_machine (main) \$ aws --version
bash: aws: command not found
- @23-22411-027-max → /workspaces/terraform_machine (main) \$ terraform --version
bash: terraform: command not found
- @23-22411-027-max → /workspaces/terraform_machine (main) \$ ansible --version
bash: ansible: command not found
- @23-22411-027-max → /workspaces/terraform_machine (main) \$

- @23-22411-027-max → /workspaces/terraform_machine (main) \$ aws sts get-caller-identity
bash: aws: command not found
- @23-22411-027-max → /workspaces/terraform_machine (main) \$

Task 1

- @23-22411-027-max → /workspaces/terraform_machine (main) \$ mkdir -p ~/.ssh
- @23-22411-027-max → /workspaces/terraform_machine (main) \$ ls ~/.ssh
- @23-22411-027-max → /workspaces/terraform_machine (main) \$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:duXmk7eJttNtxh83I9bnepyV6mDWcFx2dYI507hE5M codespace@codespaces-1de268

@23-22411-027-max → /workspaces/terraform_machine (main) \$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""

SHA256:duXmk7eJttNtxh83I9bnepyV6mDWcFx2dYI507hE5M codespace@codespaces-1de268

The key's randomart image is:

```
+--[ED25519 256]--+
|                ..*==|
|                +E**|
|                . =..|
|                o o ..|
|                S . o ...|
```

- @23-22411-027-max → /workspaces/terraform_machine (main) \$ ls -la ~/.ssh
total 20
drwxr-xr-x 2 codespace codespace 4096 Jan 8 17:30 .
drwxr-x--- 1 codespace codespace 4096 Jan 8 17:30 ..
-rw----- 1 codespace codespace 419 Jan 8 17:30 id_ed25519
-rw-r--r-- 1 codespace codespace 109 Jan 8 17:30 id_ed25519.pub
- @23-22411-027-max → /workspaces/terraform_machine (main) \$

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ ls -la ~/.ssh
drwxr-xr-x 2 codespace codespace 4096 Jan  8 17:30 .
drwxr-x--- 1 codespace codespace 4096 Jan  8 17:30 ..
-rw----- 1 codespace codespace  419 Jan  8 17:30 id_ed25519
-rw-r--r-- 1 codespace codespace  109 Jan  8 17:30 id_ed25519.pub
● @23-22411-027-max →/workspaces/terraform_machine (main) $ cd /workspaces/terraform_machine
touch terraform.tfvars
ls -la terraform.tfvars
-rw-rw-rw- 1 codespace codespace 0 Jan  8 17:32 terraform.tfvars
○ @23-22411-027-max →/workspaces/terraform_machine (main) $
```



```
@23-22411-027-max →/workspaces/terraform_machine (main) $ sudo apt update
sudo apt install -y gnupg software-properties-common curl
Setting up gpgsm (2.4.4-2ubuntu17.4) ...
Setting up dirmngr (2.4.4-2ubuntu17.4) ...
Setting up keyboxd (2.4.4-2ubuntu17.4) ...
Setting up gnupg (2.4.4-2ubuntu17.4) ...
Setting up gnupg2 (2.4.4-2ubuntu17.4) ...
Processing triggers for man-db (2.12.0-4build2) ...
> @23-22411-027-max →/workspaces/terraform_machine (main) $
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ sudo apt update
sudo apt install -y terraform
Hit:9 http://security.ubuntu.com/ubuntu noble-security InRelease
Fetched 280 kB in 1s (233 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  terraform
0 upgraded, 1 newly installed, 0 to remove and 53 not upgraded.
Need to get 30.6 MB of archives.
After this operation, 101 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com noble/main amd64 terraform amd64 1.14.3-1 [30.6 MB]
Fetched 30.6 MB in 0s (147 MB/s)
Selecting previously unselected package terraform.
(Reading database ... 58629 files and directories currently installed.)
Preparing to unpack .../terraform_1.14.3-1_amd64.deb ...
Unpacking terraform (1.14.3-1) ...
Setting up terraform (1.14.3-1) ...
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

```
Setting up terraform (1.14.3-1) ...
@23-22411-027-max →/workspaces/terraform_machine (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
@23-22411-027-max →/workspaces/terraform_machine (main) $ S
```

```

Initializing... aws/0.15.0/wheel-0.43.1.0.150-1110/WHEEL
● @23-22411-027-max →/workspaces/terraform_machine (main) $ sudo apt update
sudo apt install -y unzip
Hit:1 https://apt.releases.hashicorp.com noble InRelease
Hit:2 https://packages.microsoft.com/repos/microsoft-ubuntu-noble-prod noble InRelease
Hit:3 https://dl.yarnpkg.com/debian stable InRelease
Hit:4 https://repo.anaconda.com/pkg/misc/debrepo/conda stable InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unzip is already the newest version (6.0-28ubuntu4.1).
0 upgraded, 0 newly installed, 0 to remove and 53 not upgraded.
○ @23-22411-027-max →/workspaces/terraform_machine (main) $ S

```

```

@23-22411-027-max →/workspaces/terraform_machine (main) $ cd /workspaces/terraform_machine
terraform init

```

```

- myapp-subnet in modules/subnet
- myapp-webserver in modules/webserver
Initializing provider plugins...
- Finding latest version of hashicorp/http...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```

○ @23-22411-027-max →/workspaces/terraform_machine (main) $

```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
module.myapp-webserver[0].aws_security_group.web_sg: Creation complete after 4s [id=sg-068807f35c79]
module.myapp-webserver[1].aws_security_group.web_sg: Creation complete after 4s [id=sg-0dc9ecce8d3f]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 12s [id=subnet-01cdef74f2b3]
module.myapp-webserver[1].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Creating...
module.myapp-webserver[1].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 13s [id=i-0d62e81acbb0]
module.myapp-webserver[1].aws_instance.myapp-server: Creation complete after 14s [id=i-03594de305f5]
```

Apply complete! Resources: 10 added, 0 changed, 0 destroyed.

Outputs:

```
webserver_public_ips = [
  "158.252.77.208",
  "51.112.44.227",
]
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

```
● @23-22411-027-max →/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "158.252.77.208",
  "51.112.44.227",
]
```

Task 2

```
@23-22411-027-max → /workspaces/terraform_machine (main) $ pipx install ansible-core
```

```
ansible --version
```

```
- ansible-config
- ansible-console
- ansible-doc
- ansible-galaxy
- ansible-inventory
- ansible-playbook
- ansible-pull
- ansible-test
- ansible-vault
```

```
done! 🌟🌟🌟
```

```
ansible [core 2.20.1]
```

```
config file = None
```

```
configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
```

```
ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/site-packages/ansible
```

```
ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
```

```
executable location = /usr/local/py-utils/bin/ansible
```

```
python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/venvs/ansible-core/bin/python)
```

```
jinja version = 3.1.6
```

```
pyyaml version = 6.0.3 (with libyaml v0.2.5)
```

```
@23-22411-027-max → /workspaces/terraform_machine (main) $
```

```
pyyaml version = 6.0.3 (with libyaml v0.2.5)
```

```
@23-22411-027-max → /workspaces/terraform_machine (main) $ terraform output
```

```
webserver_public_ips = [
```

```
    "158.252.77.208",
```

```
    "51.112.44.227",
```

```
]
```

```
@23-22411-027-max → /workspaces/terraform_machine (main) $
```

```
    "51.112.44.227",
```

```
]
```

```
● @23-22411-027-max → /workspaces/terraform_machine (main) $ touch hosts
```

```
ls -la hosts
```

```
-rw-rw-rw- 1 codespace codespace 0 Jan  8 17:46 hosts
```

```
○ @23-22411-027-max → /workspaces/terraform_machine (main) $
```

```
≡ hosts
```

```
1 158.252.77.208 ansible_user=ec2-user ansible_ssh_private_key_file=/home/
```

```
2 51.112.44.227 ansible_user=ec2-user ansible_ssh_private_key_file=/home/c
```

```
3
```



```
@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[ERROR]: Task failed: Failed to connect to the host via ssh: Host key verification fail
Origin: <adhoc 'ping' task>
```

```
{'action': 'ping', 'args': {}, 'timeout': 0, 'async_val': 0, 'poll': 15}
```

```
51.112.44.227 | UNREACHABLE! => {
  "changed": false,
  "msg": "Task failed: Failed to connect to the host via ssh: Host key verification f
  "unreachable": true
}
158.252.77.208 | UNREACHABLE! => {
  "changed": false,
  "msg": "Task failed: Failed to connect to the host via ssh: Host key verification f
  "unreachable": true
}
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
```

```
covery.html for more information.
```

```
51.112.44.227 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

```
[WARNING]: Host '158.252.77.208' is using the discovered Python interpreter at '/usr/bin/python
9', but future installation of another Python interpreter could cause a different interpreter t
be discovered. See https://docs.ansible.com/ansible-core/2.20/reference\_appendices/interpreter\_
scovery.html for more information.
```

```
158.252.77.208 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

Task3

```

module "myapp-webserver" {
  public_key = var.public_key
  my_ip = local.my_ip
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_id = module.myapp-subnet.subnet.id

  # Loop count
  count = 3
  # Use count.index to differentiate instances
  instance_suffix = count.index
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash + v [] [] ... | [] [] X

```

@23-22411-027-max → /workspaces/terraform_machine (main) $ terraform apply -auto-approve
module.myapp-webserver[2].aws_instance.myapp-server: Creation complete after 13s [id=i-08c587974097b9212]

```

Apply complete! Resources: 3 added, 2 changed, 0 destroyed.

Outputs:

```

webserver_public_ips = [
  "158.252.77.208",
  "51.112.44.227",
  "51.112.53.130",
]

```

```

@23-22411-027-max → /workspaces/terraform_machine (main) $

```

terraform.tfvars hosts U X main.tf M

hosts

```

2  51.112.44.227 ansible_user=ec2-user ansible_ssh_private_key_file=/home/c
3  [ec2]
4  158.252.77.208
5  51.112.44.227
6
7  [ec2:vars]
8  ansible_user=ec2-user
9  ansible_ssh_private_key_file=/home/codespace/.ssh/id_ed25519
10 ansible_ssh_common_args='-o StrictHostKeyChecking=no'
11
12 [droplet]
13 18.212.33.101

```

Loading...

```

@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible ec2 -i hosts -m ping
}
[WARNING]: Host '158.252.77.208' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.77.208 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
○ @23-22411-027-max →/workspaces/terraform_machine (main) $

```

```

@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible 158.252.77.208 -i hosts -m ping
g
[WARNING]: Host '158.252.77.208' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.77.208 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
○ @23-22411-027-max →/workspaces/terraform_machine (main) $

```

```

@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible droplet -i hosts -m ping
@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible droplet -i hosts -m ping
[WARNING]: Host '51.112.53.130' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
51.112.53.130 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
@23-22411-027-max →/workspaces/terraform_machine (main) $

```



```

    changed : raise,
    "ping": "pong"
  }
• @23-22411-027-max → /workspaces/terraform_machine (main) $ vim ~/.ansible.cfg
• @23-22411-027-max → /workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Host '51.112.53.130' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
51.112.53.130 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '158.252.77.208' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.77.208 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  }
}

```

0 / 0 0 Ln 15, Col 1 Spaces: 4 U

terraform.tfvars hosts U X main.tf M

```

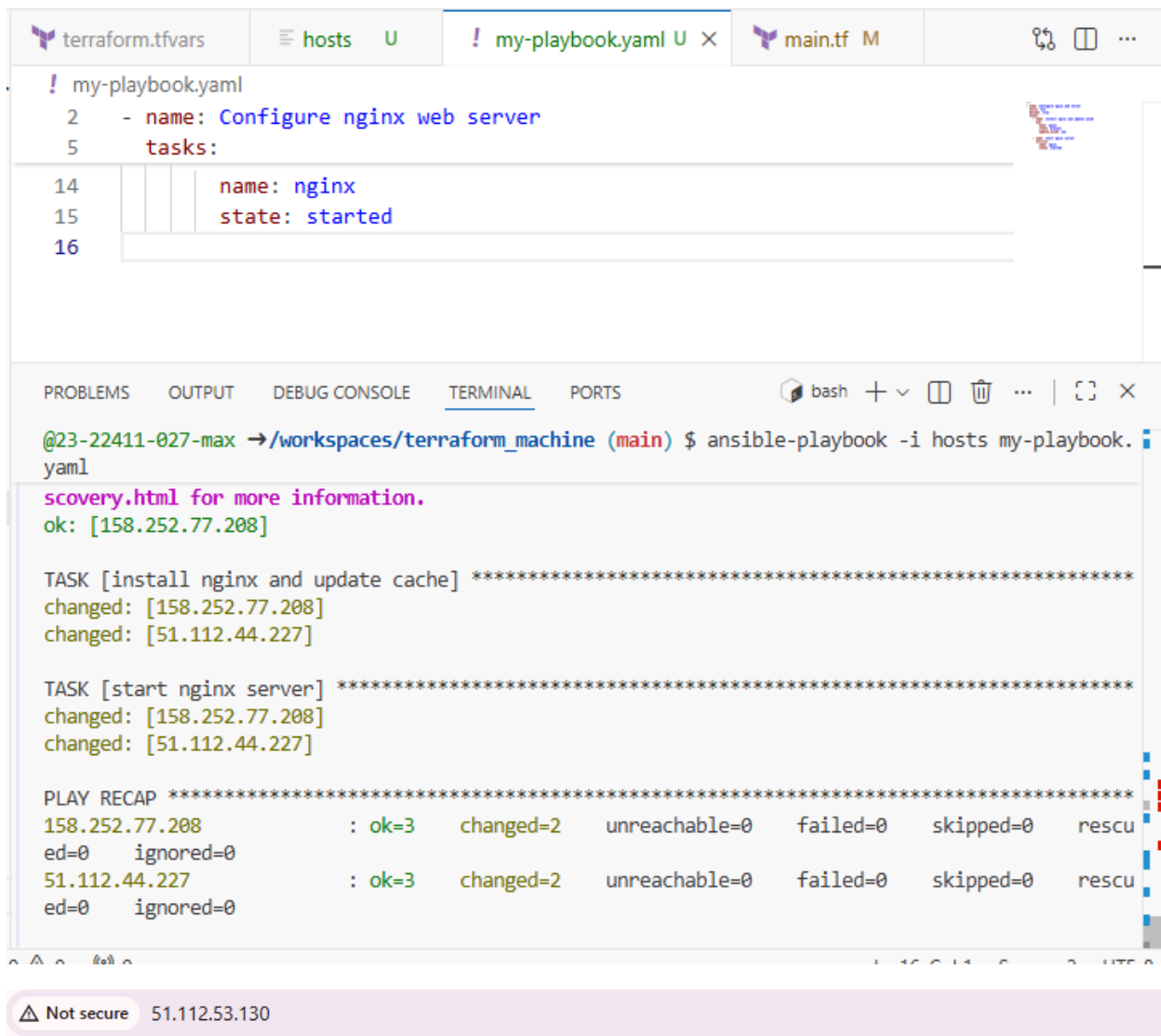
hosts
1 [ec2]
2 158.252.77.208
3 51.112.44.227
4
5 [ec2:vars]
6 ansible_user=ec2-user
7 ansible_ssh_private_key_file=/home/codespace/.ssh/id_ed25519
8
9 [droplet]
10 51.112.53.130
11
12 [droplet:vars]
13 ansible_user=ec2-user
14 ansible_ssh_private_key_file=/home/codespace/.ssh/id_ed25519
15

```

```

}
• @23-22411-027-max → /workspaces/terraform_machine (main) $ touch my-playbook.yaml
• @23-22411-027-max → /workspaces/terraform_machine (main) $ ls -la my-playbook.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 8 18:18 my-playbook.yaml
• @23-22411-027-max → /workspaces/terraform_machine (main) $

```



The screenshot shows a Visual Studio Code editor with a workspace containing several files: `terraform.tfvars`, `hosts`, `my-playbook.yaml`, and `main.tf`. The `my-playbook.yaml` file is open, showing an Ansible playbook with the following content:

```
! my-playbook.yaml
2  - name: Configure nginx web server
5    tasks:
14      - name: nginx
15        state: started
16
```

Below the editor, the `TERMINAL` panel is active, showing the output of the command `ansible-playbook -i hosts my-playbook.yaml`. The output indicates that the playbook was executed successfully on two hosts: `158.252.77.208` and `51.112.44.227`. The tasks performed were `install nginx and update cache` and `start nginx server`. The final output shows the status of the tasks for each host:

```
PLAY RECAP *****
158.252.77.208      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescue=0
                    : ignored=0
51.112.44.227      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescue=0
                    : ignored=0
```

At the bottom of the terminal, a status bar indicates that the connection is `Not secure` and the IP address is `51.112.53.130`.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
```

```
~
~
~
~
~
~
~
~
~
~
```

```
# Loop count
```

```
count = 1
```

```
# Use count.index to differentiate ins
```

```
instance_suffix = count.index
```

```
@23-22411-027-max → /workspaces/terraform_machine (main) $ terraform apply -auto-approve
```

```
module.myapp-webserver[1].aws_instance.myapp-server: Still destroying... [id=i-03594ef69fde305f5  
00m50s elapsed]
```

```
module.myapp-webserver[1].aws_instance.myapp-server: Still destroying... [id=i-03594ef69fde305f5  
01m00s elapsed]
```

```
module.myapp-webserver[1].aws_instance.myapp-server: Destruction complete after 1m1s
```

```
module.myapp-webserver[1].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-1]
```

```
module.myapp-webserver[1].aws_security_group.web_sg: Destroying... [id=sg-0dc9ed262bce8d3f]
```

```
module.myapp-webserver[1].aws_key_pair.ssh-key: Destruction complete after 0s
```

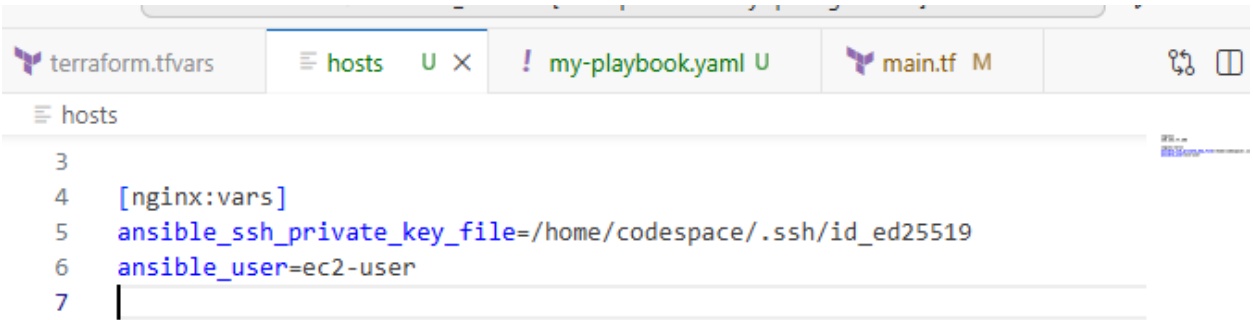
```
module.myapp-webserver[1].aws_security_group.web_sg: Destruction complete after 1s
```

```
Apply complete! Resources: 0 added, 0 changed, 6 destroyed.
```

```
Outputs:
```

```
webserver_public_ips = [  
  "158.252.77.208",  
]
```

```
@23-22411-027-max → /workspaces/terraform_machine (main) $
```



The image shows a code editor interface with four tabs at the top: 'terraform.tfvars', 'hosts', 'my-playbook.yaml', and 'main.tf'. The 'hosts' tab is currently active. Below the tabs, the text 'hosts' is displayed. The main editing area contains the following code:

```
3
4 [nginx:vars]
5 ansible_ssh_private_key_file=/home/codespace/.ssh/id_ed25519
6 ansible_user=ec2-user
7 |
```

Line numbers 3 through 7 are visible on the left side of the editor. A small cursor is positioned at the end of line 7.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

 vim + -  

```
- name: install nginx and update cache
  yum:
    name: nginx
    state: present
    update_cache: yes

- name: install openssl
  yum:
    name: openssl
    state: present

- name: start nginx server
  service:
    name: nginx
    state: started
    enabled: true
```

@23-22411-027-max →/workspaces/terraform_machine (main) \$ ansible-playbook -i hosts my-playbook.yaml

[scovery.html](#) for more information.

ok: [158.252.77.208]

TASK [install nginx and update cache] *****
ok: [158.252.77.208]

TASK [install openssl] *****
ok: [158.252.77.208]

TASK [start nginx server] *****
changed: [158.252.77.208]

PLAY RECAP *****
158.252.77.208 : ok=4 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

@23-22411-027-max →/workspaces/terraform_machine (main) \$

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task6

```
SERT --  
msg: "Public IP: {{ public_ip.content }}"  
  
name: Generate self-signed SSL certificate  
command: >  
  openssl req -x509 -nodes -days 365  
  -newkey rsa:2048  
  -keyout /etc/ssl/private/selfsigned.key  
  -out /etc/ssl/certs/selfsigned.crt  
  -subj "/CN={{ public_ip.content }}"  
  -addext "subjectAltName=IP:{{ public_ip.content }}"  
  -addext "basicConstraints=CA:FALSE"  
  -addext "keyUsage=digitalSignature,keyEncipherment"  
  -addext "extendedKeyUsage=serverAuth"  
args:  
creates: /etc/ssl/certs/selfsigned.crt
```

023-22411-027-max → /workspaces/terraform machine (1

Task7

```
./aws/dist/wheel-0.45.1.dist-info:
INSTALLER LICENSE.txt METADATA RECORD REQUESTED WHEEL direct_url.json entry_points.txt

./files:
index.php

./modules:
subnet webserver

./modules/subnet:
main.tf outputs.tf variables.tf

./modules/webserver:
main.tf outputs.tf variables.tf

./templates:
nginx.conf.j2
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

vim + ▾ □ 🗑 ... | [

```
        # PHP block
        location ~ \.php$ {
            include fastcgi_params;
            fastcgi_pass unix:/run/php-fpm/www.sock;
            fastcgi_index index.php;
            fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        }
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}
```

@23-22411-027-max → /workspaces/terraform_machine (main) \$ cat templates/nginx.conf.j2

```
        # PHP block
        location ~ \.php$ {
            include fastcgi_params;
            fastcgi_pass unix:/run/php-fpm/www.sock;
            fastcgi_index index.php;
            fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
        }
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}
```

@23-22411-027-max → /workspaces/terraform_machine (main) \$

```
template:
  src: templates/nginx.conf.j2
  dest: /etc/nginx/nginx.conf
  owner: root
  group: root
  mode: '0644'

- name: Restart nginx
  service:
    name: nginx
    state: restarted

- name: Start and enable php-fpm
  service:
    name: php-fpm
    state: started
    enabled: true
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
```

```
TASK [Copy website files] *****
ok: [158.252.77.208]
```

```
TASK [Copy nginx.conf template] *****
changed: [158.252.77.208]
```

```
TASK [Restart nginx] *****
changed: [158.252.77.208]
```

```
TASK [Start and enable php-fpm] *****
changed: [158.252.77.208]
```

```
PLAY RECAP *****
158.252.77.208      : ok=17  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Task 8

```
@23-22411-027-max → /workspaces/terraform_machine (main) $ terraform destroy -auto-approve
00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-05be332cca6877fe5
, 00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 49s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0d62e3c29681acbb0,
00m50s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 52s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-0688008a107f35c79]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-01cdeff5fdc74f2b3]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0ca74cf7f88eb25e1]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@23-22411-027-max → /workspaces/terraform_machine (main) $
```

Ln 112, Col 1 Spaces: 2 UTF-

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
module.myapp-webserver[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-06b33c1
68d7ec9]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 12s [id=subnet-04240e61
3ab886]
module.myapp-webserver[0].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 12s [id=i-0cca841
9bbaad4]
```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

```
webserver_public_ips = [
  "158.252.94.223",
]
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

```
^ 0 0
```

Ln 112, Col 1 Spaces: 2

```
o @23-22411-027-max →/workspaces/terraform_machine (main) $
```

```
● @23-22411-027-max →/workspaces/terraform_machine (main) $ terraform output
```

```
webserver_public_ips = [
  "158.252.94.223",
]
```

```
o @23-22411-027-max →/workspaces/terraform_machine (main) $
```

```
△ 0 0
```

Ln 112, Col 1

```
18 - name: Install Docker Compose
```

```
22 tasks:
```

```
23 - name: create docker cli-plugins directory
```

```
28
```

```
29 - name: install docker-compose
```

```
30 get_url:
```

```
31 url: https://github.com/docker/compose/releases/latest/download/
```

```
32 dest: /usr/local/lib/docker/cli-plugins/docker-compose
```

```
33 mode: '0755'
```

```
34
```

```
35 - name: View architecture of the system
```

```
36 debug:
```

```
37 msg: "System architecture of {{ inventory_hostname }} is {{ ansi
```

```
38
```

```
39 - name: Alternate method to view architecture of the system
```

```
40 debug:
```

```
41 msg: "System architecture of {{ inventory_hostname }} is {{ look
```

```
42
```


Task 9

```

18 - name: Install Docker Compose
22   tasks:
43     - name: restart docker service
47 - name: Adding user to docker group
48   hosts: all
49   become: true
50   vars_files:
51     - project-vars.yaml
52   tasks:
53     - name: add user to docker group
54       user:
55         name: "{{ normal_user }}"
56         groups: docker
57         append: yes

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

2 2 2 2 2 2 2

```
75 - name: Deploy Docker Containers
76 vars_files:
77     - vars.yml
78 tasks:
79     - name: check if docker-compose file exists
80       stat:
81         path: /home/{{ normal_user }}/compose.yaml
82       register: compose_file
83
84     - name: copy docker-compose file
85       copy:
86         src: "{{ docker_compose_file_location }}/compose.yaml"
87         dest: /home/{{ normal_user }}/compose.yaml
88         mode: '0644'
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

vim + v

```
- webnet

volumes:
  gitea_postgres:
    name: gitea_postgres
  gitea:
    name: gitea

networks:
  webnet:
    name: webnet

-- INSERT --
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
```

```
TASK [copy docker-compose file] *****
changed: [158.252.94.223]
```

```
TASK [deploy containers using docker-compose] *****
ok: [158.252.94.223]
```

```
PLAY RECAP *****
158.252.94.223      : ok=17   changed=3   unreachable=0   failed=0   skipped=1   rescued=0   ignored=0
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
```

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
webserver_public_ips = [
  "158.252.94.223",
]
```

```
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

0 0

Ln 1, Col 1 Spaces: 2 UTF-8

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *	PostgreSQL
Host *	db:5432
Username *	gitea
Password *	
Database Name *	gitea
SSL *	Disable
Schema	

Leave blank for database default ("public").

General Settings

Site Title *	Gitea: Git with a cup of tea
Repository Root Path *	/data/git/repositories

You can enter your company name here.

Remote Git repositories will be saved to this directory.

Tsak10

```

GNU nano 7.2                                main.tf *
instance_type = var.instance_type
availability_zone = var.availability_zone
public_key = var.public_key
my_ip = local.my_ip
vpc_id = aws_vpc.myapp_vpc.id
subnet_id = module.myapp-subnet.subnet.id

# Loop count
count          = 1
# Use count.index to differentiate instances
instance_suffix = count.index
}

resource "null_resource" "configure_server" {
  triggers = {
    webserver_public_ips_for_aws = join(",", [for i in module.myapp-webserver : i.aws_instance.public_ip])
  }

  depends_on = [module.myapp-webserver]

  provisioner "local-exec" {
    command = <<EOT
      ansible-playbook -i ${self.triggers.webserver_public_ips_for_aws}, \
        --private-key "${var.private_key}" --user ec2-user \
        my-playbook.yaml
    EOT
  }
}

```

```

@23-22411-027-max → /workspaces/terraform_machine (main) $ terraform destroy -auto-approve
00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-05be332cca6877fe5
, 00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 49s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0d62e3c29681acbb0,
00m50s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 52s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-0688008a107f35c79]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-01cdeff5fdc74f2b3]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0ca74cf7f88eb25e1]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@23-22411-027-max → /workspaces/terraform_machine (main) $

```

^ 0 0

Ln 112, Col 1 Spaces: 2 UTF-

```

tasks:
  - name: check if docker-compose file exists
    stat:
      path: /home/{{ normal_user }}/compose.yaml
    register: compose_file

  - name: copy docker-compose file
    copy:
      src: "{{ docker_compose_file_location }}/compose.yaml"
      dest: /home/{{ normal_user }}/compose.yaml
      mode: '0644'
    when: not compose_file.stat.exists

  - name: deploy containers using docker-compose
    command: docker compose up -d
    register: compose_result
    changed_when: "'Creating' in compose_result.stdout or 'Recreating' in compose_result.stdout"

- name: Wait for some time to ensure system readiness
  hosts: all
  tasks:
    - name: Wait 300 seconds for port 22 to become open and contain "OpenSSH"
      wait_for:
        port: 22
        host: "{{ inventory_hostname }}"
        delay: 10
        timeout: 300
      delegate_to: localhost

```

```

@23-22411-027-max → /workspaces/terraform_machine (main) $ terraform apply -auto-approve
module.myapp-webserver[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-06b33c68d7ec9]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 12s [id=subnet-04240e63ab886]
module.myapp-webserver[0].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 12s [id=i-0cca849bbaad4]

```

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

```

webserver_public_ips = [
  "158.252.94.223",
]

```

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *	PostgreSQL
Host *	db:5432
Username *	gitea
Password *	
Database Name *	gitea
SSL *	Disable
Schema	

Leave blank for database default ("public").

General Settings

Site Title *	Gitea: Git with a cup of tea
You can enter your company name here.	
Repository Root Path *	/data/git/repositories
Remote Git repositories will be saved to this directory.	

Task 11

Setting up terraform (1.14.3-1) ...

- @23-22411-027-max → /workspaces/lab14 (main) \$ terraform --version
Terraform v1.14.3
on linux_amd64
 - @23-22411-027-max → /workspaces/lab14 (main) \$
- 0

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS nano - task11_dynamic_inventory
GNU nano 7.2 ansible.cfg *
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
deprecation_warnings = False

enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519
█
```

```
OFF 1110X 00000000
@23-22411-027-max →/workspaces/lab14 (main) $ mkdir task11_dynamic_inventory
@23-22411-027-max →/workspaces/lab14 (main) $ cd task11_dynamic_inventory
@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $ nano ansible.cfg
@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $ touch inventory_aws_ec2.yaml
@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $ █

@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $ ls -la inventory_aws_ec2.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 10 18:32 inventory_aws_ec2.yaml
@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $ █
```

```
GNU nano 7.2 inventory_aws_ec2.yaml *
---
plugin: aws_ec2
regions:
  - me-central-1
█
```



```
GNU nano 7.2 main.tf *
ami          = "ami-0c02fb55956c7d316"
instance_type = "t3.nano"
subnet_id    = aws_subnet.myapp_subnet.id

tags = {
  Name = "dev-webserver"
  env  = "dev"
}

resource "aws_instance" "prod" {
  ami          = "ami-0c02fb55956c7d316"
  instance_type = "t3.nano"
  subnet_id    = aws_subnet.myapp_subnet.id

  tags = {
    Name = "prod-webserver"
    env  = "prod"
  }
}
```

...

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS



```
GNU nano 7.2 outputs.tf
output "webserver_public_ips" {
  value = aws_instance.dev.public_ip
}

output "prod-webserver_public_ips" {
  value = aws_instance.prod.public_ip
}
```

```
● @23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
○ @23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $ terraform apply -a
```

```
... PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - task11_dynamic_inventory + v [] ...

@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $ terraform apply -auto-approve
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ primary_network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ prod-webserver_public_ips = (known after apply)
+ webserver_public_ips      = (known after apply)
aws_instance.prod: Creating...
aws_instance.dev: Creating...
aws_instance.prod: Still creating... [00m10s elapsed]
aws_instance.dev: Still creating... [00m10s elapsed]
aws_instance.prod: Creation complete after 13s [id=i-0a3b4735d7cf43050]
aws_instance.dev: Creation complete after 13s [id=i-0531fd9536fff082e]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
prod-webserver_public_ips = "51.112.253.167"
webserver_public_ips = "3.29.244.31"
@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $ terraform output
prod-webserver_public_ips = "51.112.253.167"
webserver_public_ips = "3.29.244.31"
@23-22411-027-max →/workspaces/lab14/task11_dynamic_inventory (main) $
```

```
webservers_public_ips = "3.29.244.31"
@23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $ terraform output
prod-webservers_public_ips = "51.112.253.167"
webservers_public_ips = "3.29.244.31"
@23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $ python3 -m pip install boto3 botocore
Collecting boto3
  Downloading boto3-1.42.25-py3-none-any.whl.metadata (6.8 kB)
Collecting botocore
  Downloading botocore-1.42.25-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3)
  Downloading jmespath-1.0.1-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/codespace/.local/lib/python3.12/site-packages
s (from botocore) (2.9.0.post0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/codespace/.local/lib/python3.12/site-packages
(from botocore) (2.5.0)
Requirement already satisfied: six>=1.5 in /home/codespace/.local/lib/python3.12/site-packages (from python-date
util<3.0.0,>=2.1->botocore) (1.17.0)
Downloading boto3-1.42.25-py3-none-any.whl (140 kB)
Downloading botocore-1.42.25-py3-none-any.whl (14.6 MB)
 14.6/14.6 MB 27.4 MB/s 0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.42.25 botocore-1.42.25 jmespath-1.0.1 s3transfer-0.16.0
@23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $
```



```

#---
#plugin: amazon.aws.aws_ec2
#regions:
# - me-central-1

---
plugin: amazon.aws.aws_ec2
regions:
  - me-central-1
filters:
  instance-state-name: running
keyed_groups:
  - key: tags.Env
    prefix: env

---
plugin: aws_ec2
regions:
  - me-central-1

keyed_groups:
  - key: tags
    prefix: tag
    separator: "_"

```

```

$ ansible-inventory -i inventory_aws_ec2.yaml --graph
@all:
  |--@ungrouped:
  |--@aws_ec2:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
  |--@tag_env_dev:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |--@tag_Name_dev_webserver:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |--@tag_Name_prod_webserver:
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
  |--@tag_env_prod:
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com

$ nano inventory_aws_ec2.yaml
$ ansible-inventory -i inventory_aws_ec2.yaml --graph
@all:
  |--@ungrouped:
  |--@aws_ec2:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
  |--@tag_env_dev:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |--@tag_Name_dev_webserver:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |--@tag_Name_prod_webserver:
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
  |--@tag_env_prod:
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com

```

```

(venv) @23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $ nano inventory_aws_ec2.yaml
(venv) @23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
@all:
  |--@ungrouped:
  |--@aws_ec2:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
  |--@tag_env_dev:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |--@tag_Name_dev_webserver:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |--@instance_type_t3_nano:
  |   |--ec2-3-29-244-31.me-central-1.compute.amazonaws.com
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
  |--@tag_Name_prod_webserver:
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
  |--@tag_env_prod:
  |   |--ec2-51-112-253-167.me-central-1.compute.amazonaws.com
(venv) @23-22411-027-max → /workspaces/lab14/task11_dynamic_inventory (main) $

```

```

%0

```

```

-i inventory_aws_ec2.yaml \
-u ec2-user \
--private-key ~/.ssh/id_ed25519 \
-l instance_type_t3_micro \

```

```

PLAY [Deploy Docker Containers] *****

TASK [Gathering Facts] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [check if docker-compose file exists] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [copy docker-compose file] *****
skipping: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [deploy containers using docker-compose] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

PLAY [Wait for some time to ensure system readiness] *****

TASK [Gathering Facts] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [Wait 300 seconds for port 22 to become open and contain "OpenSSH"] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com -> localhost]

PLAY RECAP *****
ec2-40-172-101-68.me-central-1.compute.amazonaws.com : ok=17  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0

```



```

ansible-playbook \
-i inventory_aws_ec2.yaml \
-l instance_type_t3_micro \
-u ec2-user \ --

PLAY [Deploy Docker Containers] *****

TASK [Gathering Facts] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [check if docker-compose file exists] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [copy docker-compose file] *****
skipping: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [deploy containers using docker-compose] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

PLAY [Wait for some time to ensure system readiness] *****

TASK [Gathering Facts] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com]

TASK [Wait 300 seconds for port 22 to become open and contain "OpenSSH"] *****
ok: [ec2-40-172-101-68.me-central-1.compute.amazonaws.com -> localhost]

PLAY RECAP *****
ec2-40-172-101-68.me-central-1.compute.amazonaws.com : ok=17  changed=1  unreachable=0  failed=0  skipped=2  rescued=0  ignored=0

```

Task 13

```

GNU nano 7.2                                ansible/ansible.cfg *
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3

GNU nano 7.2                                inventory/hosts *
[nginx]
40.172.101.68

[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user

```



```
roles/ssl/tests:  
inventory test.yml
```

```
roles/ssl/vars:  
main.yml
```

```
roles/webapp:  
README.md defaults files handlers meta tasks templates tests vars
```

```
roles/webapp/defaults:  
main.yml
```

```
roles/webapp/files:
```

```
roles/webapp/handlers:  
main.yml
```

```
roles/webapp/meta:  
main.yml
```

```
roles/webapp/tasks:  
main.yml
```

```
roles/webapp/templates:
```

```
roles/webapp/tests:  
inventory test.yml
```

```
---
```

```
GNU nano 7.2 roles/nginx/handlers/main.yml *  
#SPDX-License-Identifier: MIT-0  
---  
# handlers file for nginx  
- name: Restart nginx  
  service:  
    name: nginx  
    state: restarted
```

```
GNU nano 7.2
#SPDX-License-Identifier: MIT-0
```

```
roles/nginx/tasks/main.yml *
```

```
---
# tasks file for nginx
- name: Install nginx
  yum:
    name: nginx
    state: present
    update_cache: yes
  notify: Restart nginx

- name: Install openssl
  yum:
    name: openssl
    state: present

- name: Start and enable nginx
  service:
    name: nginx
    state: started
    enabled: true
```

```
GNU nano 7.2
```

```
my-playbook.yml *
```

```
---
- name: Deploy NGINX Web Stack
  hosts: nginx
  become: true
  roles:
    - nginx
```

```
GNU nano 7.2
```

```
roles/ssl/defaults/main.yml *
```

```
#SPDX-License-Identifier: MIT-0
---
# defaults file for ssl
imds_v2_token_ttl: "3600"
ssl_days_valid: 365
```

GNU nano 7.2

roles/ssl/tasks/main.yml *

```
- name: Save public IP as fact
  set_fact:
    server_public_ip: "{{ public_ip.content }}"

- name: Generate self-signed certificate
  command: >
    openssl req -x509 -nodes -days {{ ssl_days_valid }}
    -newkey rsa:2048
    -keyout /etc/ssl/private/selfsigned.key
    -out /etc/ssl/certs/selfsigned.crt
    -subj "/CN={{ server_public_ip }}"
    -addext "subjectAltName=IP:{{ server_public_ip }}"
  args:
    creates: /etc/ssl/certs/selfsigned.crt
```

GNU nano 7.2

roles/webapp/defaults/main.yml *

```
#SPDX-License-Identifier: MIT-0
---
# defaults file for webapp
nginx_user: nginx
nginx_worker_processes: auto
nginx_worker_connections: 1024
nginx_error_log_level: notice

web_root: /usr/share/nginx/html
web_index_file: index.php
```

GNU nano 7.2

roles/webapp/files/index.php *

```
<div class="info"><span class="label">Hostname:</span> <?= htmlspecialchars($hostname) ?></div>
<div class="info"><span class="label">Instance ID:</span> <?= htmlspecialchars($instance_id) ?></div>
<div class="info"><span class="label">Private IP:</span> <?= htmlspecialchars($private_ip) ?></div>
<div class="info"><span class="label">Public IP:</span> <?= htmlspecialchars($public_ip) ?></div>
<div class="info"><span class="label">Public DNS:</span>
  <a href="https://<?= htmlspecialchars($public_dns) ?>" target="_blank">
    https://<?= htmlspecialchars($public_dns) ?></a>
</div>
<div class="info"><span class="label">Deployed:</span> <?= $deployed_date ?></div>
<div class="info"><span class="label">Status:</span> ☒ Active and Running</div>
<div class="info"><span class="label">Managed By:</span> Terraform + Ansible</div>
</div>
</body>
</html>
```

```
GNU nano 7.2
#SPDX-License-Identifier: MIT-0
---
# handlers file for webapp
- name: Restart nginx
  service:
    name: nginx
    state: restarted

- name: Restart php-fpm
  service:
    name: php-fpm
    state: restarted
```

```
GNU nano 7.2 roles/webapp/t
- name: Install PHP packages
  yum:
    name:
      - php-fpm
      - php-curl
    state: present
  notify: Restart php-fpm

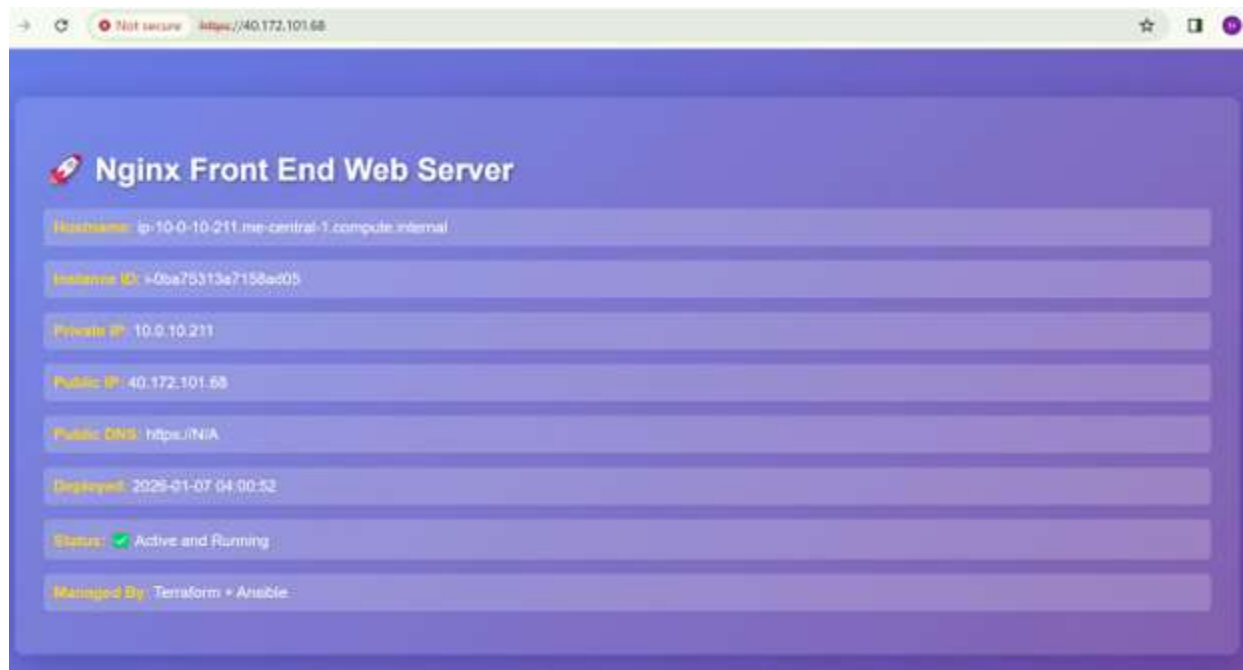
- name: Copy PHP website
  copy:
    src: index.php
    dest: "{{ web_root }}/{{ web_index_file }}"
    owner: nginx
    group: nginx
    mode: '0644'
  notify: Restart nginx

- name: Deploy nginx config
  template:
    src: nginx.conf.j2
    dest: /etc/nginx/nginx.conf
  notify: Restart nginx

- name: Start and enable php-fpm
  service:
    name: php-fpm
    state: started
    enabled: true
```

```
GNU nano 7.2 my-playbook.yaml *
---
- name: Deploy NGINX Web Stack
  hosts: nginx
  become: true
  roles:
    - nginx

---
- name: Deploy NGINX Web Stack with SSL and PHP
  hosts: nginx
  become: true
  roles:
    - nginx
    - ssl
    - webapp
```



Cleanup:

```
@23-22411-027-max →/workspaces/terraform_machine (main) $ terraform destroy -auto-approve
00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-05be332cca6877fe5
, 00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 49s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0d62e3c29681acbb0,
00m50s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 52s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-0688008a107f35c79]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-01cdeff5fdc74f2b3]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0ca74cf7f88eb25e1]
aws_vpc.myapp_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 7 destroyed.

```
@23-22411-027-max →/workspaces/terraform_machine (main) $
```

0 0

Ln 112, Col 1 Spaces: 2 UTF-

terraform.tfstate X

terraform.tfstate

```
1 {
2   "version": 4,
3   "terraform_version": "1.14.3",
4   "serial": 81,
5   "lineage": "777679b8-fda0-89a1-cfd0-1694d76e5efd",
6   "outputs": {
7     "webserver_public_ips": {
8       "value": [
9         "3.28.136.127"
10      ],
11      "type": [
12        "tuple",
13        "string"
14      ]
15    }
16  }
17 }
```